

the mind discards false assumptions, or a time in which solutions to similar problems are retrieved from memory, among other ideas.

Each theory can be represented mathematically in artificial intelligence models – indeed, they bear a resemblance to the computational creativity guidelines set down by Simon, Newell and Shaw. However, most models choose between theories rather than seeking to incorporate multiple theories and therefore are fragmentary at best.

Sun and Hèlie's Explicit-Implicit Interaction (EII) theory integrates several of the competing theories into a larger equation.

“EII unifies a lot of fragmentary pre-existing theories,” Sun said. “These pre-existing theories only account for some aspects of creative problem solving, but not in a unified way. EII unifies those fragments and provides a more coherent, more complete theory.”

The basic principles of EII propose the coexistence of two different types of knowledge and processing: explicit and implicit. Explicit knowledge is easier to access and verbalise, can be rendered symbolically, and requires more attention to process. Implicit knowledge is relatively inaccessible, harder to verbalise, is more vague and requires less attention to process.

In solving a problem, explicit knowledge could be the knowledge used in reasoning, deliberately thinking through different options, while implicit knowledge is the intuition that gives rise to a solution suddenly. Both types of knowledge are involved simultaneously to solve a problem and reinforce each other in the process. By including this principle in each step, Sun was able to achieve a successful system.

“This tells us how creative problem solving may emerge from the interaction of explicit and implicit cognitive processes; why both types of processes are necessary for creative problem solving, as well as in many other psychological domains and functionalities,” said Sun.

Creativity and art in Artificial Intelligence

The idea of creativity is usually linked with another intrinsically human quality: Imagination. Indeed, when it comes to art, the two seem almost intertwined. And there can be no question about art's usefulness. It's no longer an arbitrary and meaningless pursuit, as several psychologists and philosophers have spoken about its therapeutic and inspirational qualities.

But does it fit into the concept of artificial intelligence? Well, yes and no. It depends on how you want to look at the field of AI. According